

AV_Culpability_E6_transparency boundary_Study4

Start of Block: informed_consent

ic Informed Consent

Participation is voluntary

It is your choice whether or not to participate in this research. If you choose to participate, you may change your mind and leave the study at any time. Refusal to participate or stopping your participation will involve no penalty or loss of benefits to which you are otherwise entitled.

What is the purpose of this research?

The purpose of this research is to examine human judgments about automated vehicles. All data from this experiment are gathered for scientific purposes and will contribute to our eventual understanding of brain and visual function. These data may be published in scientific journals so that other researchers may have access to these data.

How long will I take part in this research?

Your participation will take approximately 10 minutes to complete.

What can I expect if I take part in this research?

As a participant, you will be asked to look at images presented on a video display and give responses with key presses or movements of a mouse pointing device. Your response may involve responding as quickly as you can, memorizing what you saw, making a judgment, or completing a questionnaire. You will also be asked to complete a demographics form.

What are the risks and possible discomforts?

If you choose to participate, the effects should be comparable to those you would experience from viewing a computer monitor for 10 minutes and using a mouse or keyboard, e.g., eye fatigue. You are free to take breaks throughout the session.

Will I be compensated for participating in this research?

You will be compensated \$1.60 for this study. You will still receive payment if you withdraw early.

If I take part in this research, how will my privacy be protected? What happens to the information you collect?

Your participation in this experiment will remain confidential, and your identity will not be stored with your data.

If I have any questions, concerns or complaints about this research study, who can I talk to?

The researcher for this study is Julian De Freitas who can be reached at 626.559.6401; #161 Morgan Hall, 15 Harvard Way, Boston MA, 02163; jdefreitas@hbs.edu. If you have questions, concerns, or complaints, If you would like to talk to the research team, If you think the research has harmed you, or If you wish to withdraw from the study. This research has been reviewed by the Committee on the Use of Human Subjects in Research at Harvard University. They can be reached at 617-496-2847, 1350 Massachusetts Avenue, Suite 935, Cambridge, MA 02138, or cuhs@harvard.edu for any of the following: If your questions, concerns, or complaints are not being answered by the research team, If you cannot reach the research team, If you want to talk to someone besides the research team, or If you have questions about your rights as a research participant.

☐ I agree (1)

☐ I disagree (2)

End of Block: informed_consent

Start of Block: prolific ID



id What is your Prolific ID?

Please note that this response should auto-fill with the correct ID.

End of Block: prolific ID

Start of Block: attention_checks

att_1 **This is an attention check.**

John is taller than Paul. Who is shorter?

- ☐ John (1)
 - ☐ Paul (2)
 - ☐ Neither John nor Paul (3)
 - ☐ Both John and Paul (4)
-

att_2 **What color is grass? The fresh, uncut grass, not leaves or hay.**

Please make sure to select purple, so that we know you're paying attention.

- ☐ Green (1)
- ☐ Purple (2)

End of Block: attention_checks

Start of Block: attention check fail

att_fail Unfortunately, you have failed to answer at least one of the attention check questions correctly and have been redirected to the end of the survey.
Please click the button to below to be redirected back to Prolific and register your submission.

End of Block: attention check fail

Start of Block: av_general

av_info **Please read the information presented below carefully in order to be able to understand and answer the survey questions**

Below is a table depicting 6 possible types of vehicles. The different types are based on how many driving tasks are controlled by the human driver ('HUMAN') versus the self-driving system ('MACHINE').

In level 1, the human controls all driving tasks, whereas in level 6, the machine controls all

driving tasks. The levels in-between differ in which driving tasks are controlled by the human versus the machine.

End of Block: av_general

Start of Block: comp_checks

comp_1 **You were shown a table. What did the table represent?**

- ☐ The number of typical daily tasks controlled by humans vs. machines (1)
 - ☐ Different types of vehicles based on how many driving tasks are controlled by humans vs. machines (2)
 - ☐ Six people's preferences for humans vs. machines (3)
-

comp_2 **What were the values for vehicle type #1 (the one at the top of the table)?**

- ☐ MACHINE, HUMAN, HUMAN, MACHINE (1)
- ☐ HUMAN, MACHINE, MACHINE, HUMAN (2)
- ☐ MACHINE, MACHINE, MACHINE, MACHINE (3)
- ☐ HUMAN, HUMAN, HUMAN, HUMAN (4)

End of Block: comp_checks

Start of Block: av_frame1

frame1_info1 AEON is an automobile firm that is planning on introducing a new AV program in the market. They have named their program: **AEON AUTOPILOT**.

Please answer some questions so that AEON can understand and analyze preferences for AVs and investigate the potential of their AV program.

Page Break

auto_1 What do you perceive as the level of automation for AEON's AUTOPILOT Program?

☐ 1 (1)

☐ 2 (2)

☐ 3 (3)

☐ 4 (4)

☐ 5 (5)

☐ 6 (6)

frame1_info2

End of Block: av_frame1

Start of Block: av_frame1_acc

frame1_info3 Now, please imagine the following scenario involving **Aeon's Autopilot**:

A man is being driven home by his car's Autopilot system, when he gets a text message on his phone and very quickly looks down to check it.

At that point, neither he nor the Autopilot system notices the Stop sign. The car drives right past it and smashes into another car in front of it, injuring the driver of that car.

frame1_info4 As it turns out, **Aeon's Autopilot** has been classified as **Level 2 automation** by regulators.

This information CANNOT be found on the company's website but CAN be found on NHTSA's (National Highway Traffic and Safety Administration) website.

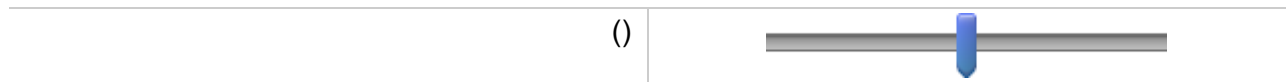
For your reference, here is the table again that shows which driving tasks are managed by the human driver versus the driving system at different levels of automation.

frame1_info5 **Please rate the extent to which you agree with the following statements:**

resp_software1 **Aeon's Autopilot** is responsible for the accident.

Completely disagree Neither agree nor disagree Completely agree

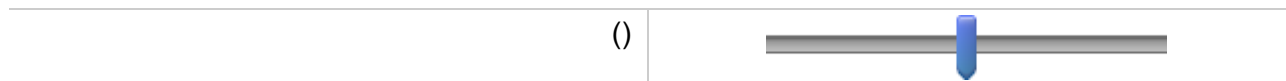
0 10 20 30 40 50 60 70 80 90 100



resp_human1 **The human in the car** is responsible for the accident.

Completely disagree Neither agree nor disagree Completely agree

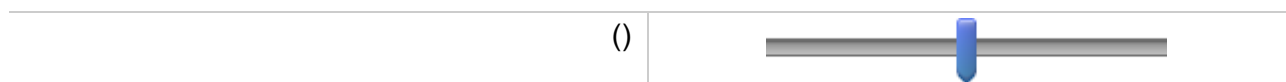
0 10 20 30 40 50 60 70 80 90 100



liable_firm1 Aeon, **the company**, is liable for the damages from the accident.

Completely disagree Neither agree nor disagree Completely agree

0 10 20 30 40 50 60 70 80 90 100



liable_human1 **The human in the car** is liable for the damages from the accident.

Completely
disagree Neither agree
nor disagree Completely
agree

0 10 20 30 40 50 60 70 80 90 100

()



liab_text1 In a sentence or two, please explain why you think either Aeon, **the company**, or the **human driver** is more liable for the accident?

End of Block: av_frame1_acc

Start of Block: av_frame2

frame2_info1 AEON is an automobile firm that is planning on introducing a new AV program in the market. They have named their program: **AEON COPILOT**.

Please answer some questions so that AEON can understand and analyze preferences for AVs and investigate the potential of their AV program.

Page Break

co_1 What do you perceive as the level of automation for AEON's COPILOT Program?

☐ 1 (1)

☐ 2 (2)

☐ 3 (3)

☐ 4 (4)

☐ 5 (5)

☐ 6 (6)

frame2_info2

End of Block: av_frame2

Start of Block: av_frame2_acc

frame2_info3 Now, please consider the following scenario involving **Aeon's Copilot**:

A man is being driven home by his car's Copilot system, when he gets a text message on his phone and very quickly looks down to check it.

At that point, neither he nor the Copilot system notices the Stop sign. The car drives right past it and smashes into another car in front of it, injuring the driver of that car.

frame2_info4 As it turns out, **Aeon's Copilot** has been classified as **Level 2 automation** by regulators.

This information CANNOT be found on the company's website but CAN be found on NHTSA's (National Highway Traffic and Safety Administration) website.

For your reference, here is the table again that shows which driving tasks are managed by the human driver versus the driving system at different levels of automation.

frame2_info5 **Please rate the extent to which you agree with the following statements:**

resp_software2 Aeon's **Copilot** is responsible for the accident.

Completely disagree Neither agree nor disagree Completely agree

0 10 20 30 40 50 60 70 80 90 100

()



resp_human2 **The human in the car** is responsible for the accident.

Completely disagree Neither agree nor disagree Completely agree

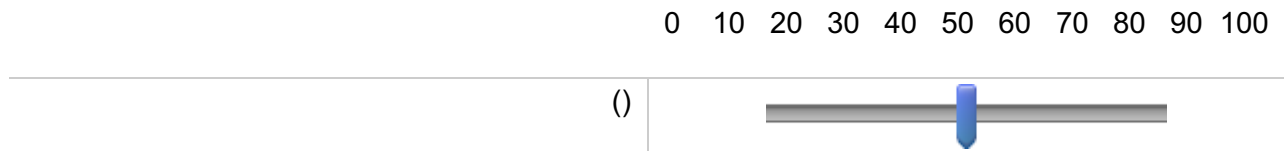
0 10 20 30 40 50 60 70 80 90 100

()

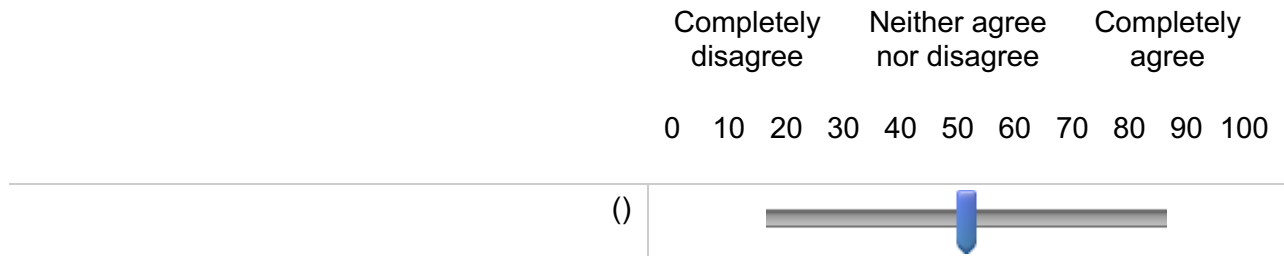


liab_firm2 Aeon, **the company**, is liable for the damages from the accident.

Completely disagree Neither agree nor disagree Completely agree



liab_human2 The **human in the car** is liable for the damages from the accident.



liab_text2 In a sentence or two, please explain why you think either Aeon, **the company**, or the **human driver** is more liable for the accident?

End of Block: av_frame2_acc

Start of Block: av_frame3

frame3_info1 AEON is an automobile firm that is planning on introducing a new AV program in the market. They have named their program: **AEON AUTOPILOT**.

Please answer some questions so that AEON can understand and analyze preferences for AVs and investigate the potential of their AV program.

Page Break

auto_2 What do you perceive as the level of automation for AEON's AUTOPILOT Program?

- ☐ 1 (1)
 - ☐ 2 (2)
 - ☐ 3 (3)
 - ☐ 4 (4)
 - ☐ 5 (5)
 - ☐ 6 (6)
-

frame3_info2

End of Block: av_frame3

Start of Block: av_frame3_acc

frame3_info3 Now, please imagine the following scenario involving **Aeon's Autopilot**:

A man is being driven home by his car's Autopilot system, when he gets a text message on his phone and very quickly looks down to check it.

At that point, neither he nor the Autopilot system notices the Stop sign. The car drives right past it and smashes into another car in front of it, injuring the driver of that car.

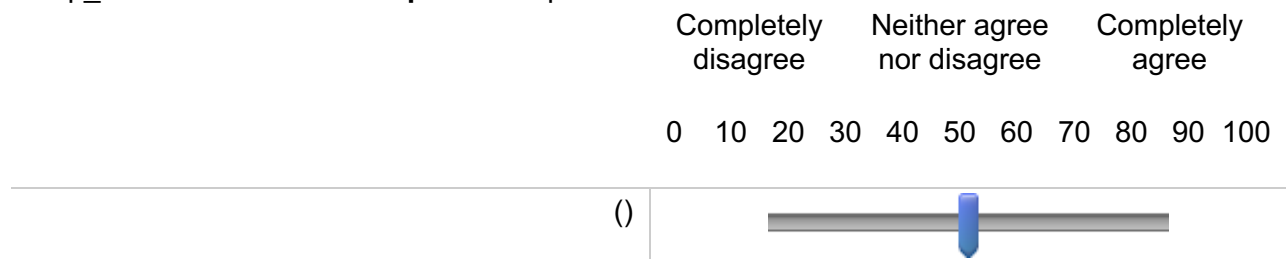
frame3_info4 As it turns out, **Aeon's Autopilot** has been classified as **Level 2 automation** by regulators.

This information CAN be found on the company's website and CAN also be found on NHTSA's (National Highway Traffic and Safety Administration) website.

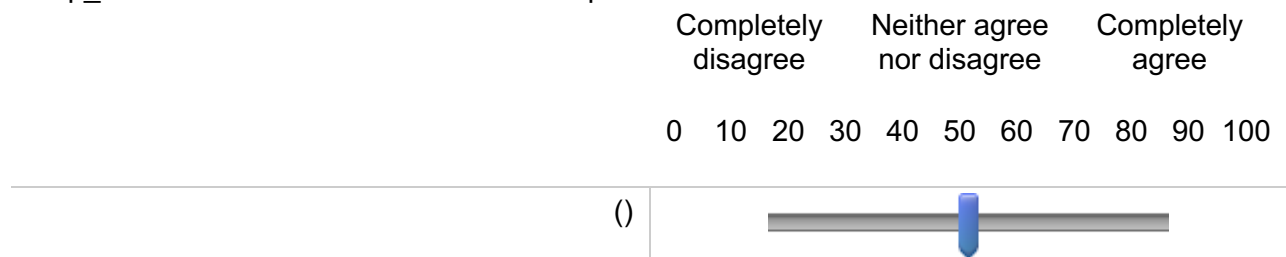
For your reference, here is the table again that shows which driving tasks are managed by the human driver versus the driving system at different levels of automation.

frame3_info5 **Please rate the extent to which you agree with the following statements:**

resp_software3 **Aeon's Autopilot** is responsible for the accident.

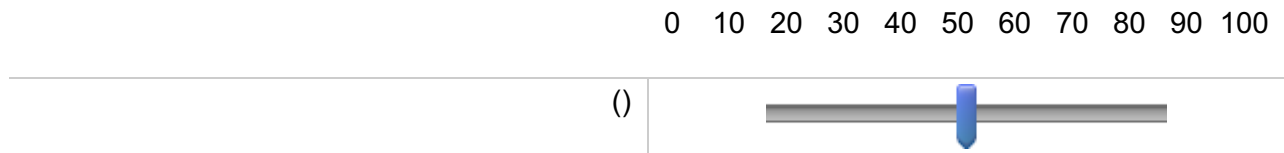


resp_human3 **The human in the car** is responsible for the accident.

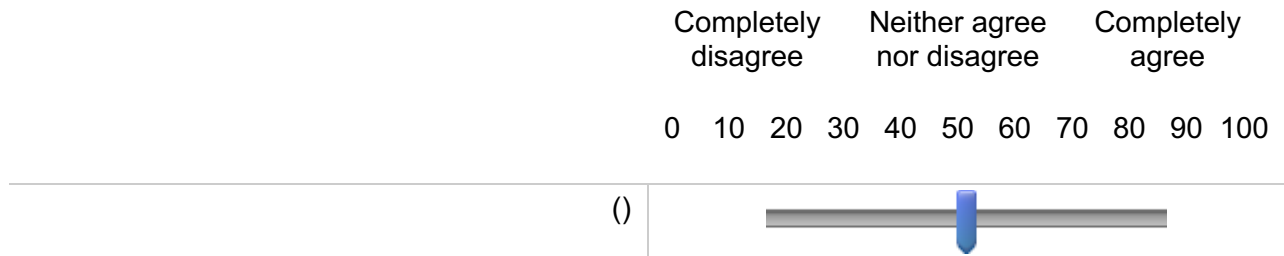


liab_soft3 Aeon, **the company**, is liable for the damages from the accident.

Completely disagree	Neither agree nor disagree	Completely agree
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liab_human3 **The human in the car** is liable for the damages from the accident.



liab_text3 In a sentence or two, please explain why you think either Aeon, **the company**, or the **human driver** is more liable for the accident?

End of Block: av_frame3_acc

Start of Block: av_frame4

frame4_info1 AEON is an automobile firm that is planning on introducing a new AV program in the market. They have named their program: **AEON COPILOT**.

Please answer some questions so that AEON can understand and analyze preferences for AVs and investigate the potential of their AV program.

Page Break

co_2 What do you perceive as the level of automation for AEON's COPILOT Program?

- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)

frame4_info2

End of Block: av_frame4

Start of Block: av_frame4_acc

frame4_info3 Now, please imagine the following scenario involving **Aeon's Copilot**:

A man is being driven home by his car's Copilot system, when he gets a text message on his phone and very quickly looks down to check it.

At that point, neither he nor the Copilot system notices the Stop sign. The car drives right past it and smashes into another car in front of it, injuring the driver of that car.

frame4_info4 As it turns out, **Aeon's Copilot** has been classified as **Level 2 automation** by regulators.

This information CAN be found on the company's website and CAN also be found on NHTSA's (National Highway Traffic and Safety Administration) website.

For your reference, here is the table again that shows which driving tasks are managed by the human driver versus the driving system at different levels of automation.

frame4_info5 **Please rate the extent to which you agree with the following statements:**

resp_software4 **Aeon's Copilot** is responsible for the accident.

Completely disagree Neither agree nor disagree Completely agree

0 10 20 30 40 50 60 70 80 90 100

()



resp_human4 **The human in the car** is responsible for the accident.

Completely disagree Neither agree nor disagree Completely agree

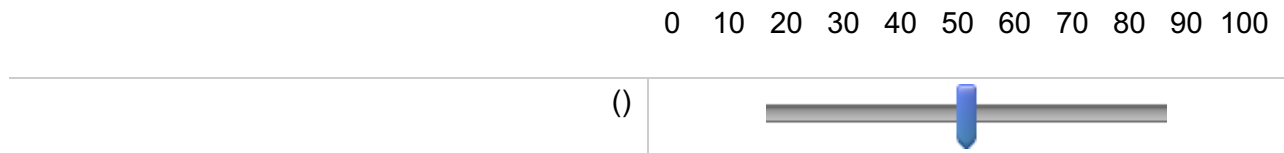
0 10 20 30 40 50 60 70 80 90 100

()



liab_software4 Aeon, **the company**, is liable for the damages from the accident.

Completely disagree Neither agree nor disagree Completely agree



liab_human4 **The human in the car** is liable for the damages from the accident.

Completely disagree Neither agree nor disagree Completely agree

0 10 20 30 40 50 60 70 80 90 100



liab_text4 In a sentence or two, please explain why you think either Aeon, **the company**, or the **human driver** is more liable for the accident?

End of Block: av_frame4_acc

Start of Block: comp_checks2

comp_3 Based on the information provided to you, indicate the level of automation assigned to Aeon's system by regulators.

- ☐ Level 2 automation (1)
 - ☐ Level 3 automation (2)
 - ☐ Level 4 automation (3)
 - ☐ Level 5 automation (4)
 - ☐ Level 6 automation (5)
-

comp_4 Which of the following is true about information regarding this level of automation?

- ☐ It CANNOT be found on the company's website (2)
- ☐ It CAN be found on the company's website (3)

End of Block: comp_checks2

Start of Block: demographics

gender What is your gender?

- ☐ Male (1)
 - ☐ Female (2)
 - ☐ Prefer not to disclose (3)
 - ☐ Other (please specify) (4)
-

ethnicity What is your ethnicity?

- ☐ Black or African American (1)
- ☐ Asian (2)
- ☐ White or European (3)
- ☐ Hispanic or Latino (4)
- ☐ Mixed (5)
- ☐ Prefer Not to Say (7)
- ☐ Other (please specify) (6)

age What is your age (in years)?

education Please indicate the highest level of education completed.

- ☐ High School or Equivalent (1)
- ☐ Vocational/Technical School (2 year) (2)
- ☐ Some College (3)
- ☐ College Graduate (4 year) (4)
- ☐ Masters Degree (MS) (5)
- ☐ Doctoral Degree (PhD) (6)
- ☐ Professional Degree (MD, JD, etc.) (7)
- ☐ Other (please specify) (8)
-

license **Do you have a driving license?**

- ☐ Yes (1)
- ☐ No (2)
-

ai_knowledge **How much would you say you know about automated cars?**

Very little

A lot

0 10 20 30 40 50 60 70 80 90 100

()



End of Block: demographics

Start of Block: end

Q144 Thank you for taking part in the survey. Please click the button to below to be redirected back to Prolific and register your submission.

End of Block: end
